L*(96) = 8, and it is DEFINED.

For the first time, L* rises with reasoning depth.

At rho=0.60, with realized density identical at 0.594 in every bin, the ladder clears the 0.90 gate in 0 of 6 depth bins at L=4, 3 of 6 at L=6, and 6 of 6 at L=8. So the threshold is 8 -- and the three bins L=6 fails are the three DEEPEST ones (T=7, 8, 9), which is the shape finding 11 never saw.

Finding 11 measured L*(T) across 20 bins and found it flat: +0.00 layers per round, L*=3 at every T. Those bins were all at N=24, where T spans 1-5 and every cell is either floored or saturated. At N=96, T spans 4-9 and L=6 sits ON the threshold -- the only place the effect can show. It shows there and nowhere else: the trend is significant only at L=6.

Leaked-order run (shuffle_adj=False), so every number here is a DECODING score, not a deduction score (findings 16-17). The shared-pool best is still L=12 at 0.9273; L=8 ties it (0.9113, delta -0.0160, p=0.091) at 1.5x fewer layers.

job 49639312, COMPLETED 17:45:48, ExitCode 0:0 | scripts/slurm/sudoku2_n96_layer_scan_80k.sbatch

What the model actually sees

Every token is one categorical id into a 106-symbol vocabulary (N=96, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING BEGIN_ROUND SOLVED UNRESOLVED	*section + round markers*
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3	*the q colours*
10-105	N0 N1 ... N95	*one id per node, no structure in the id*

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*
SOLVED	

The graph section is a neighbour table, not an image: node, then its degree neighbours. THE ORDER WITHIN A ROW IS THE LEAK of finding 16.

THIS INSTANCE, 777 TOKENS (seed 50000000002)

HINTS	BEGIN_HINTS N2 COLOR_1 N3 COLOR_2 N4 COLOR_1 N5 COLOR_3 N6 COLOR_1 N7 COLOR_2 ... (129 tokens)
GRAPH	BEGIN_GRAPH N0 N67 N84 N34 N49 N59 N1 N86 N4 N17 N53 N44 ... (577 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N0 COLOR_2 N11 COLOR_2 N16 COLOR_0 N20 COLOR_0 N35 COLOR_1 N37 ... (71 tokens)

Loss is on the REASONING section only: 5 rounds, 32 deductions, 64 target tokens. Everything before BEGIN_REASONING is the problem statement.

One real reasoning trace

64 hints of 96 nodes, then 5 rounds of simultaneous deduction. Hints: N2=1 N3=2 N4=1 N5=3 N6=1 N7=2 N8=3 N9=3 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND        N0 COLOR_2 N11 COLOR_2 N16 COLOR_0 N20 COLOR_0 ...
  18 deduced
BEGIN_ROUND        N1 COLOR_2 N13 COLOR_1 N15 COLOR_2 N19 COLOR_3 ...
  9 deduced
BEGIN_ROUND        N42 COLOR_2 N46 COLOR_0 N93 COLOR_1
  3 deduced
BEGIN_ROUND        N88 COLOR_3
  1 deduced
... 1 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N0, neighbours [67, 84, 34, 49, 59]

colours known before this round:

```
   [0, 1, 3]  (3 = q-1)
```

only colour left: 2

```
so the trace emits  N0 COLOR_2
```

The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

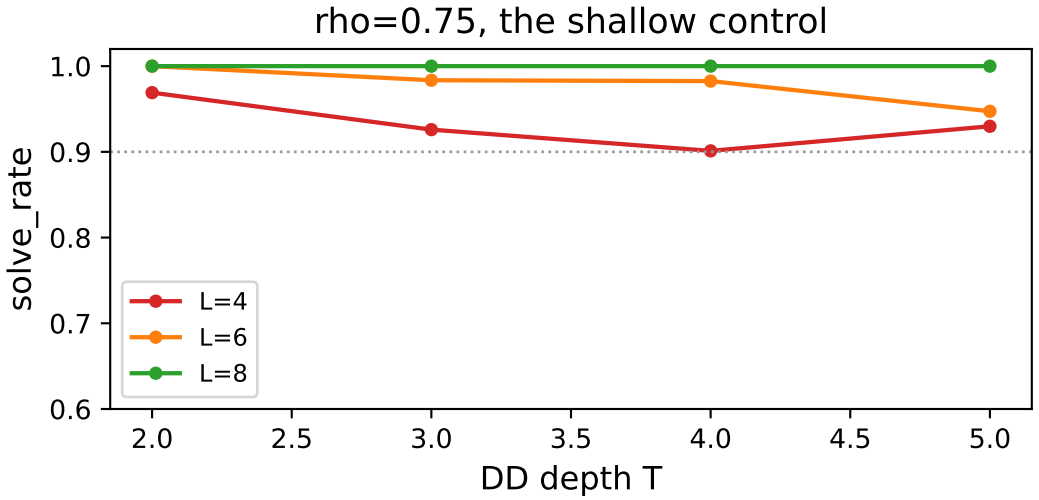
T=5 rounds, the median of 40 draws from this config (range 1-19); seed 50000000002, shuffle_adj=False; evidence from sudoku_tokens.evidence_rounds (pre-round state, as DD saw it)

The ladder, bin by bin

L* is the smallest num_layers that reaches 0.90 on instances of DD depth T at FIXED density.

	T=4	T=5	T=6	T=7	T=8	T=9	bins>=0.90	val_nll
L=4	0.756	0.801	0.734	0.681	0.765	0.744	0/6	1.4983
L=6	0.919	0.925	0.902	0.884	0.824	0.884	3/6	1.4203
L=8	0.953	0.932	0.930	0.942	0.941	0.930	6/6	1.4141

n per bin: T=4:86, T=5:146, T=6:143, T=7:69, T=8:68, T=9:43 (555 instances per row, realized density 0.594 in every bin)



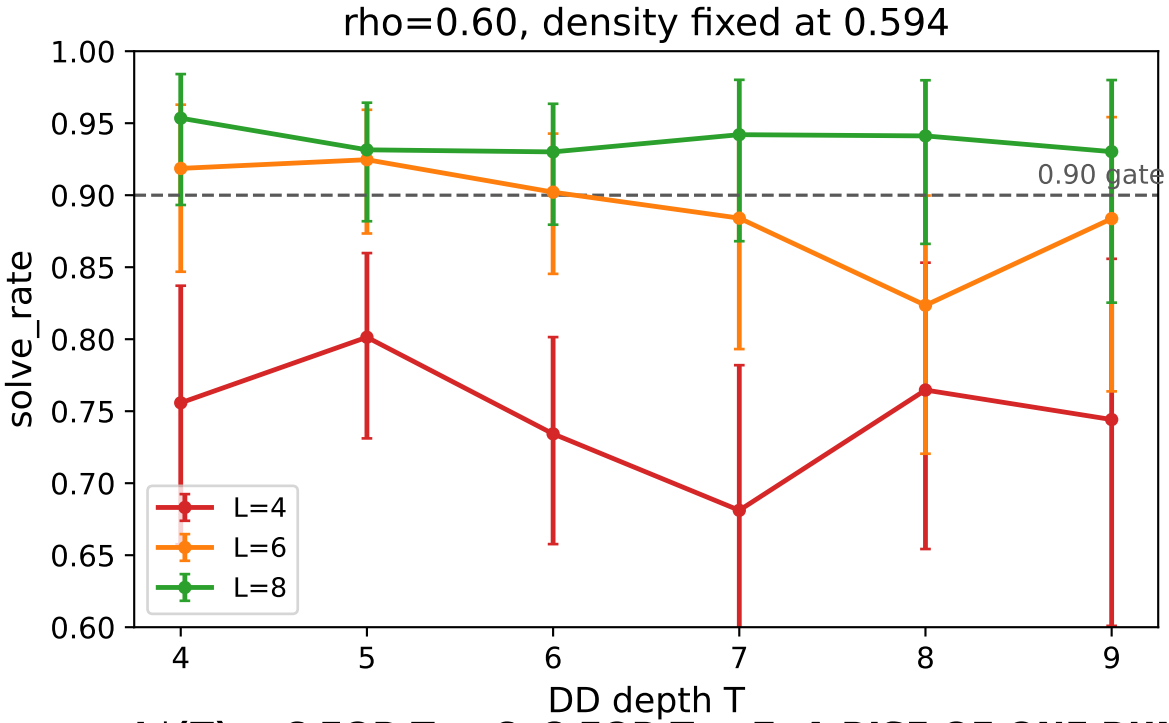
WHY THE SHALLOW CONTROL MATTERS

At rho=0.75 every cell including L=4 clears the gate at every T, and L=8 scores 1.000 in all four bins. The ladder is therefore not measuring a model that is simply bad: the same three checkpoints are at ceiling on easier instances. Whatever L=4 and L=6 lose at rho=0.60 they lose to the combination of density and depth, not to competence.

sudoku_depth_eval --rho 0.75 0.60; all three cells at n_train=80000, 150 epochs

Where L* actually moves

Only L=6 is near the gate, and only L=6 has a depth trend. That is the whole result.



	pooled	CA z	CA p	T<=6	T>=7	Fisher p
L=4	418/555 = 0.7532	-0.80	0.4232	0.7653	0.7278	0.3454
L=6	498/555 = 0.8973	-1.92	0.0546	0.9147	0.8611	0.0720
L=8	520/555 = 0.9369	-0.23	0.8144	0.9360	0.9389	1.0000

L*(T) = 6 FOR T<=6, 8 FOR T>=7: A RISE OF ONE RUNG.

Against finding 11's +0.00 layers per round over 20 bins at N=24. Not established: p=0.055 and p=0.072 are marginal, from one seed, and the two tests are not independent. The claim is narrower -- the shape finding 11 looked for appears as soon as the measurement has a cell at the threshold and a T range wide enough to split.

THE PATTERN IS THE EVIDENCE

L=4 is on the floor (0.75) and L=8 is above the gate everywhere (0.937), so neither CAN show a depth trend -- and neither does (p=0.42, p=0.81). L=6 sits at 0.897, within a hair of the gate, and it is the one cell with a trend (CA p=0.055) and a shallow-vs-deep gap (+0.054, p=0.072). A depth cost that appears only where the threshold is, is what a real threshold effect looks like.

That is also a warning for the planned leak-free L*(T) ladder at N=24: T spans 2-5 there, which is the range that produced the flat answer. It may need N=48 to be informative.

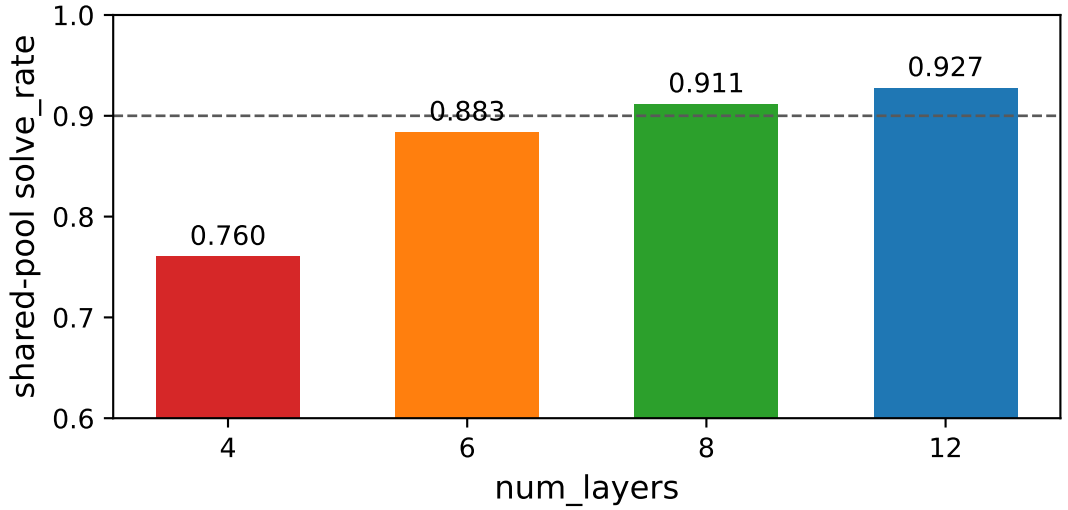
error bars are 95% Jeffreys intervals; Cochran-Armitage is the same trend test finding 11 reported at p=0.088

L* across problem size, and the cost of the threshold

All at d_model=256, ff=1024 -- finding 14 showed L* moves when the MLP narrows, so the width is part of the claim.

problem	L* at ff=1024	measured in	best solve_rate	best cell
N=24	3	finding 11/14	0.9960	L=12
N=48	4	finding 14 (49509466/49523504)	0.9573	L=8, ff=256
N=96	8	THIS RUN (49639312)	0.9273	L=12

L* = 3, 4, 8 for N = 24, 48, 96. Mean DD depth grows with N, so this is the across-problem version of the same claim the T bins make within N=96 -- and the two now agree for the first time.



THRESHOLD IS NOT OPTIMUM -- pre-registration (d)

```
On the shared n=1500 pool, paired against cell B (L=12):
  L=4   0.7600   -0.1673  p=1.7e-42
  L=6   0.8833   -0.0440  p=1.1e-05
  L=8   0.9113   -0.0160  p=0.091
  L=12  0.9273   reference

I pre-registered L=8 >= L=12. It is a TIE, not a win: the
point estimate favours L=12 by 0.016 and the test cannot
resolve it. So the threshold (8) is below the optimum (12)
but nothing above 8 is resolvably better -- which is the
separation the pre-registration was for, reached by the
weaker of the two routes.
```

sudoku_agg_eval, shared pool at seed_start=5e10, n=1500, paired McNemar exact; N=96 pool is its own instance set

Pre-registrations, as resolved

Written into scripts/slurm/sudoku2_n96_layer_scan_80k.sbatch before the job ran.

	claim	verdict	evidence
(a)	L*(96) >= 6	CONFIRMED	L=4 clears 0 of 6 bins; L* = 8
(b)	L*(96) is DEFINED inside {4, 6, 8}	CONFIRMED	L=8 clears 6 of 6 bins at rho=0.60
(c)	L*(T) is flat in T, as in finding 11	FALSIFIED (marginally)	L=6 clears T=4,5,6 and fails T=7,8,9; CA p=0.055, shallow-vs-deep 0.861 vs 0.915, Fisher p=0.072. Only at L=6.
(d)	L=8 >= L=12, separating the threshold from the optimum	TIE	0.9113 vs 0.9273, delta -0.0160, p=0.091 -- not resolvable either way

(c) is the one worth being careful about: it is the prediction I most wanted to be wrong.

I pre-registered flatness because that is what finding 11 measured, and a result that agrees with the prediction you wanted to fail deserves more scepticism than one that does not. Both tests are marginal (0.055, 0.072), they are not independent, and this is one seed. The reason I report it as falsified rather than unresolved is the pattern across cells: the trend appears at the one layer count near the gate and at neither of the two that are pinned, which is a prediction flatness does not make.

my record on pre-registrations in this project: 2 confirmed, 1 falsified, 1 tie, on this job

What this changes, and what it does not

WHAT IT CHANGES

- L*(96) was UNDEFINED after finding 15 (the control stalled at 0.8267 and cleared no bin). It is now 8, so the N ladder reads L* = 3, 4, 8 at N = 24, 48, 96.
- Finding 11's flat L*(T) is demoted from 'depth does not track depth' to 'not resolvable at N=24': at N=96, with a cell at the gate and T spanning 4-9, it rises by one rung.
- The N=96 question is no longer 'can it learn at all' but 'how deep must it be'.

WHAT IT DOES NOT

- This is a LEAKED-ORDER run (shuffle_adj=False). Every solve_rate here is a decoding score. Finding 16 showed this exact family of checkpoints collapses to 0.03-0.15 when adj[v] is permuted, so none of these numbers is a deduction score, and L* measured on a shortcut task need not equal L* on the real one.
- p=0.055 and p=0.072 are marginal, from one seed, on two non-independent tests.
- L* depends on the MLP width (finding 14): every L* above is 'at ff=1024'. A thin-deep N=96 cell could move it, and has not been run.

WHAT I WOULD DO NEXT

1. The leak-free L*(T) ladder, but at N=48 rather than only N=24 -- this run is the argument for it. N=24's T range is 2-5, the range that produced the flat answer.
2. L=10 at N=96 would tighten L* from 'between 6 and 8' to a single rung, and is one cell.
3. A thin-deep N=96 cell (ff=256), which finding 13 says should match L=8/ff=1024 at roughly a third of the parameters.